

Moni Kumari

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Professional Summary

Third-year Computer Science & Engineering (Data Science) student building AI, backend, and payments systems, with hands-on experience in **Razorpay integrations, webhook security, reconciliation, and idempotency**. Built **RetryRight**, an AI-assisted failed-payment recovery engine shortlisted for the **Razorpay AI Buildathon**, where LLMs interpret failures while deterministic policies authorise financial actions. Also experienced in shipping full-stack products, evaluating AI systems, and building data-intensive software.

Technical Skills

- **Languages:** TypeScript, Python, JavaScript, SQL
- **Backend & Payments:** Node.js, Express.js, REST APIs, Razorpay, Webhooks, Signature Verification, Idempotency, Payment Reconciliation
- **AI:** LLM APIs, Vercel AI SDK, Structured Outputs (Zod), Multimodal Vision, Evaluation & Benchmarking
- **Data & Tools:** PostgreSQL, SQLite, Firestore, MongoDB, Git, CI/CD, Vitest

Education

CMR Institute of Technology

Bachelor of Engineering – Computer Science & Engineering (Data Science), CGPA: 8.86

Bengaluru, India

2024 – 2028

Projects

RetryRight – AI Failed-Payment Recovery for Razorpay Subscriptions

[GitHub](#)

- Built an AI-assisted failed-payment recovery engine in TypeScript where an LLM classifies gateway failures while **seven deterministic policy gates** independently authorise retries based on payment state, mandate status, retry limits, economics, and budget.
- **99.5% classification accuracy across 208 payments with 0 unsafe actions**; a confidently misclassified revoked mandate was blocked by an independent authorisation gate, resulting in **₹0 loss from model error**.
- Added execution-time state revalidation, append-only audit logs, **37 tests**, adversarial testing, ablation, sensitivity analysis and **40-run variance testing**; cut inference calls from **1,040 to 145** through caching and deduplication.

UniTrack – SaaS Attendance Platform with Razorpay Billing

[Live](#) | [GitHub](#)

- **Stack:** React Native (Expo), TypeScript, Firebase, **Razorpay**, Cloud Functions, Gemini API
- **Launched to 20+ active users with paying subscribers**, processing recurring subscription payments through Razorpay with **cryptographic webhook signature verification** and **idempotent handling of duplicate callbacks**.
- Built **payment reconciliation and failed-transaction recovery** with grace periods, keeping entitlements consistent with what a customer paid for.
- Applied an **AI vision workflow** converting uploaded images into structured JSON, cutting onboarding from **15 minutes to 30 seconds**.

Quarry – Columnar Analytics Query Engine

[GitHub](#)

- Built a columnar SQL engine from scratch in Python/NumPy with a lexer, parser, query planner and vectorized execution engine - **5× faster than SQLite** on 1M rows.
- Profiled a GROUP BY bottleneck and added dictionary encoding, cutting execution **731ms to 53ms (13.8×)**; **differential testing against SQLite** across 17 query shapes caught 2 semantic bugs. 68 tests passing.

Open Source & Achievements

- **The-Boring-Education/TBE-Web** - authored a PR, passed maintainer review, shipped to production in a TypeScript/Next.js monorepo.
- **Razorpay AI Buildathon 2026** - Shortlisted for Technical Interviews.
- **Google Solution Challenge 2026** - SahayakAI, multilingual AI emergency triage (Gemini 2.5 Flash, 12 Indian languages, sub-2s).
- **INNOVATEX 4.0**, Presidency University - built **2 working prototypes in 9 hours**.